Exercise 3

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Data Science class

Student number: 2805000621
SQL case statement

1. Select Product_name, Price,
Case
When Price > 1000 then 'Expensive'
When price between 100 and 1000 then 'Mid-range'
when price < 100 then 'Budget'
End as Price_category
From Products;

	Product_name	Price	Price_category
1	Laptop	1200.00	Expensive
2	Phone	800.00	Mid-range
3	Keyboard	45.00	Budget
4	Monitor	300.00	Mid-range
5	Mouse	25.00	Budget

2. Select Customer_name, amount,
Case
When amount >= 1000 then 'High value'
When amount between 500 and 999.99 then 'Medium value'
when amount < 500 then 'Low value'
End as Order_value_category
From orders;

	Customer_name	amount	Order_value_category
1	Alice	150.00	Low value
2	Bob	560.00	Medium value

3	Charlie	999.99	Medium value
4	Diana	45.50	Low value
5	Ethan	1200.00	High value

3. Select emp_name, department, salary,
Case
when department = 'IT' and salary > 80 000
then 'Senior IT'
when department = 'HR' and salary > 55000
then 'Experienced HR'
Else 'staff'
End as position_level
From employees;

	Emp_name	department	Salary	Position_level
1	John	IT	85 000	Senior IT
2	Sara	HR	60 000	Experienced HR
3	Mark	IT	75 000	Staff
4	Lucy	Finance	95 000	Staff
5	Tom	HR	55000	Staff

4. Select Student_name, score,
Case
when score >= ~~A~~ 90 then 'A'
when score between 80 and 89 then 'B'
when score between 70 and 79 then 'C'
when score between 60 and 69 then 'D'
when score ~~between~~ < 60 then 'F'
End as grade
from students;

	Student_name	Score	Grade
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ella	68	D

5. Select delivery_id, delivery_time_minutes,
Case
When delivery_time_minutes <= 30 then 'Fast'
when delivery_time_minutes between 31 and 60 then 'On Time'
when delivery_time_minutes > 60 then 'Late'
~~From~~ End as performance from deliveries

	Delivery_id	delivery_time_min	Performance
1	1	45	On time
2	2	80	Late
3	3	30	Fast
4	4	65	Late
5	5	100	Late

6. Select Issue_type, Priority,
Case
When Priority = 3 then 'High'
when Priority = 2 then 'Medium'
when Priority = 1 then 'Low'
End as priority_label
from tickets;

	Issue_type	Priority	Priority_Level
1	Login issue	1	Low
2	Server down	3	High
3	Slow system	2	Medium
4	Email error	2	Medium
5	Password reset	1	Low

7. Select student_id, (days_present X 100 / total days) as attendence_percentage,
Case
When (days_present x 100 / total days) >= 90 then 'excellent'
When (days_present x 100 / total days) between 75 and 89 then 'Good'
When (days_present x 100 / total day) < 75 then 'needs improvement'
End as attendance_status.
from attendance;

	student_id	attendance_%	attendence_status
1	1	90	Excellent
2	2	60	Needs improvement
3	3	96	Excellent
4	4	50	Needs Improvement
5	5	100	Excellent

8. Select product_id, stock_qty,
Case
When stock_qty = 0 then 'out of stock'
When stock_qty between 1 and 5 then 'Low stock'

When stock_qty > 5 then 'In stock'
End as stock_status
From Products_inventory;

	Product_id	Stock_qty	Stock_status
1	1	5	Low stock
2	2	0	out of Stock
3	3	25	In stock
4	4	10	In stok
5	5	3	Low stock

9. Select Subject, enrolled_students,
Case
When enrolled_students >= 25 then 'Large'
when enrolled_students between 10 and 25 then 'Medium'
when enrolled_students < 10 then 'small'
End as class_size_category
From classes;

	Subject	Enrolled_Students	Class_size_category
1	Math	30	Large
2	English	25	Large
3	Science	15	Medium
4	Art	5	small
5	History	20	Medium

10. Select Payment_id, Payment_method, amount,
Case
When Payment_method = 'cash' and amount >= 200

then 'Eligible for discount'
Else 'Not Eligible'
End as discount_eligibility
from Payments;

	Payment_Id	amount	Payment_Method	discount Eligibility
1	1	50.00	Card	Not Eligible
2	2	200.00	cash	Eligible for dis
3	3	150.00	card	Not eligible
4	4	75.00	PayPal	Not eligible
5	5	300.00	Cash	Eligible for dis